Simulation and Modeling

Lecture 07

Random Number Generation

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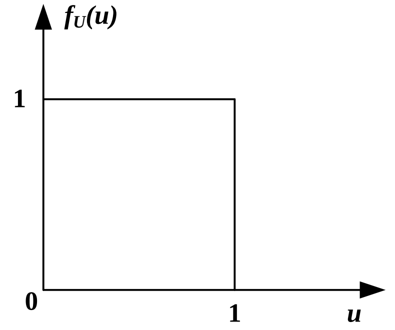
Random Numbers (1/2)

- A sequence of numbers $u_1, u_2, \ldots$ is called random numbers, if it poses two statistical properties
  - Uniformity
  - Independence
- Each number $u_i, i = 1, 2, \ldots$ is
  - Independent sample drawn from continuous uniform distribution [0, 1]
  - The PDF is

$$f_U(u) = \begin{cases} 1, & 0 \le u \le 1 \\ 0, & \text{otherwise} \end{cases}$$



Random Numbers (2/2)

- For each number $u_i, i = 1, 2, \ldots$ (contd.)
  - The expected value of the u_i's is

$$E[U] = \mu_U = \int_0^1 u.\, f_U(u) du = \frac{u^2}{2}\Big|_0^1 = \frac{1}{2}$$

  - The variance of the u_i's are

$$Var[U] = \int_0^1 (u - \mu_U)^2.\, f_U(u) du = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

- If the interval $[0, 1]$ is divided into n classes of equal length, then uniformity and independence ensures
  - If number of observations is N then
    - Expected number of observations in each class is $\frac{N}{n}$
    - Probability of observing a value in a particular class is independent of previous observations

Pseudo-Random Numbers

- Random numbers generated by known methods
  - Numbers can be replicated
  - Numbers are not truly random
    - Pseudo-random number
    - Random numbers generated by computer programs/arithmetic methods
- Pseudo-random numbers departs from ideal randomness, e.g.,
  - May not be uniformly distributed
  - May be discrete valued
    - Instead of continuous valued
  - The mean may too high or too low
  - The variance may be too high or too low
  - There may be dependence
    - Autocorrelation between numbers
    - Numbers successively higher/lower than adjacent numbers
    - Several numbers above the mean followed by several numbers below the mean

Pseudo-Random Numbers - Considerations

- Methods used for generating random numbers should be
  - Fast, i.e., computationally and storage efficient
  - Portable to different computers, or even to different programming languages
  - With sufficiently long cycle
    - Length of a sequence before the numbers repeat
  - Able to replicate the random numbers
  - Able to generate separate streams
  - Able to approximate the statistical properties of random numbers
    - Uniformity
    - Independence
    - Maximum density

Random-Number Generations

Midsquare Method (1/2)

- Proposed by von Neuman and Metropolis (1940)
- Start with a four digit positive integer Z_0
- Square Z_0 to obtain an 8-digit number
  - If required, append zeros to the left
- Take the middle 4 digits as Z_1
- Place a decimal point at the left of Z_1 to get the first random number $U(0,1), u_1$
- Z_2 be middle 4 digit of Z_1^2 and u_2 be Z_2 with a decimal point at the left

i	Z_i	u_i	Z_i^2
0	7182	-	51581124
1	5811	0.5811	33767721
2	7677	0.7677	58936329
3	9363	0.9363	87665769
4	6657	0.6657	44315649

Midsquare Method (2/2)

- Problem with the midsquare method
  - Strong tendency to degenerate fairly rapidly to zero
    - If $Z_0 = 1009$, then after few numbers it reaches to 0
  - A fundamental objection is that it is not random
    - Next number can be easily generated
    - Though this is applicable to all arithmetic generators

Linear Congruential Generators (LCG)

- A sequence of integers $Z_i, i = 1, 2, \ldots$ is defined by the recursive formula

$$Z_i = (aZ_{i-1} + c)(\text{mod}\, m)$$

  - a is the multiplier
  - m is modulus
  - c is the increment
  - Z_0 is the starting value, known as the seed

$$Z_i = (aZ_{i-1} + c) - \left\lfloor \frac{aZ_{i-1} + c}{m} \right\rfloor \times m$$

It can also be shown that z_i can be calculated directly

$$z_i = \left[a^i Z_0 + \frac{c\left(a^i - 1\right)}{a - 1} \right] (\text{mod}\, m)$$

Linear Congruential Generators (LCG)

- The random number is calculated as

$$u_i = \frac{Z_i}{m}$$

- u_i's can take on only the rational values from

$$\{0, \frac{1}{m}, \frac{2}{m}, \frac{3}{m}, \ldots, \frac{m-1}{m}\}$$

- u_i's depend on
  - Floating point operations
  - Choice of a, c, m, Z_0

LCG – Example (1/2)

- Consider the LCG defined by
  - $m = 16, a = 5, c = 3, Z_0 = 7$
  - $Z_1 = (5 \cdot 7 + 3)(\text{mod } 16) = 38 \text{ mod } 16 = 6$
  - $Z_2 = (5 \cdot 6 + 3)(\text{mod } 16) = 33 \text{ mod } 16 = 1$
  - $Z_3 = (5 \cdot 1 + 3)(\text{mod } 16) = 8 \text{ mod } 16 = 8$
  - $Z_4 = (5 \cdot 8 + 3)(\text{mod } 16) = 43 \text{ mod } 16 = 11$

LCG – Example (2/2)

i	Z_i	u_i	i	Z_{09i}	u_i
0	7	–	10	9	0.563
1	6	0.375	11	0	0.000
2	1	0.063	12	3	0.188
3	8	0.500	13	2	0.125
4	11	0.688	14	13	0.813
5	10	0.625	15	4	0.250
6	5	0.313	16	7	0.438
7	12	0.750	17	6	0.375
8	15	0.938	18	1	0.063
9	14	0.875	19	8	0.500

LCG – Example

- Observations:
  - Due to modulus operation, there can be at most m distinct Z_is
  - After a certain number, generator repeats one of the generated numbers
    - The sequence is repeated then
  - # of distinct numbers generated is called the <u>cycle length</u> (or <u>period</u>)
  - If the cycle length is m, the generator is said to have <u>full cycle</u>

Properties of Full Cycle Generators

- Every integer between 0 and $m - 1$ occurs only once
- May ensure the uniformity of u_i
  - May be non-uniform for a segment
  - Large gap for a sequence
- Full cycle depends on the careful selection of $m, a,$ and c

Theorem: Full Cycle LCG

- The LCG defined by

$$Z_i = (aZ_i + c)(\mathrm{mod}\, m)$$

has full cycle if and only if

a) Only +ve integer that divides both m and c is 1
   - gcd of m and c is 1
   - c is relatively prime to m

b) If q is a prime, and divides m, then q divides $a - 1$

c) For $k = 0, 1, 2, \ldots,$

$$a = 4k + 1$$

   - If 4 divides m, then 4 divides $a - 1$

LCG: Choice of m (1/2)

- A large m ensures long period
- A large m ensures high density
- A reasonable choice is $m = 2^{b-1}$
  - b is the word length of the computer
  - For 32-bit word, $m = 2^{31}$
    - Leftmost bit word is the sign bit
- Choosing $m = 2^b$, avoids explicit division

LCG: Choice of m(2/2)

- For 3-bit decimal operation
  - $m = 10^b = 10^2 = 100, a = 19, c = 49, Z_0 = 63$
  - $Z_0 = 63$
  - $Z_1 = (19 \times 63 + 49) \bmod 100 = 1246 \bmod 100 = 46$
  - $Z_2 = (19 \times 46 + 49) \bmod 100 = \quad 923 \bmod 100 = 23$
  - $Z_3 = (19 \times 23 + 49) \bmod 100 = \quad 486 \bmod 100 =86$
- Explicit division is avoided by integer overflow
  - Speed and efficiency
- After getting $aZ_i + c$, to obtain Z_{i+1}
  - Drops the leftmost digits, and
  - Takes only the rightmost b digits

LCG: Choice of c and a

- Choice of c
  - When $m = 2^b$ (an even number)
    - An odd number for c will make it relatively prime to m

- Choice of a
  - $a - 1$ should be divisible by 4

Multiplicative Generator

- If c=0, then the LCG is called a multiplicative generator
- Advantages: addition of c is not necessary
- Challenges:
  - Cannot have full period
    - Condition (a) not satisfied
    - However, possible to obtain a period of $m - 1$
  - $m = 2^b$ can be used to avoid explicit division
    - Cycle length is at most $2^{b-2} = \frac{m}{4}$, if
    - Z_0 is odd
    - a is of the form, for $k = 0, 1, 2, \ldots$
      - $a = 8k + 3$
      - $a = 8k + 5$
  - Not known, where the $\frac{m}{4}$ integers will fall
    - Might result in large gap
    - Non-uniformity

Multiplicative LCG

- a = 13
- m = 2^b = 64
- Z_0=1,2,3,4

i	z_i	z_i	z_i	z_i
0	1	2	3	4
1	13	26	39	52
2	41	18	59	36
3	21	42	63	20
4	17	34	51	4
5	29	58	23	
6	57	50	43	
7	37	10	47	
8	33	2	35	
9	45		7	
10	9		27	
11	53		31	
12	49		19	
13	61		55	
14	25		11	
15	5		15	
16	1		3	

Multiplicative Generator: choice of m

- Instead $m = 2^b$, m be the largest prime less than 2^b
  - For example, if $b = 31$, then
    - $m = 2^{31} - 1$
  - Cycle of $m - 1$ is achievable , if
    - a is a primitive element modulo m
      - The smallest integer l for which $a^l - 1$ is divisible by m is
      - $l = m - 1$
    - This is called prime modulus multiplicative LCG (PMMLCG)

Challenges with PMMLCG

- How can one obtain primitive element modulo m?
- Since $m \neq 2^b$, how can we avoid the division operation?

More General Congruence

- $Z_i = g(Z_{i-1}, Z_{i-2}, \ldots)(\mathrm{mod}\ m)$
  - $g(\cdot)$ is a deterministic function of previous Z_i's
  - Still the Z_i's are between $0 \sim\ m - 1$
  - $u_i = \frac{Z_i}{m}$
  - For LCG, $g(\cdot) = aZ_{i-1} + c$

Quadratic Congruential Generator

- $g(\cdot) = a'Z_{i-1}^2 + aZ_{i-1} + c$

Multiple recursive Generator

- $g(\cdot) = a_1 Z_{i-1} + a_2 Z_{i-2} + \cdots + a_q Z_{i-q}$
  - Cycle is as large as $m^q - 1$

Composite Generator

- Takes two or more separate generators
- Combine them to generator the final random number
- Expected to exhibit a longer cycle and better statistical properties